

Mixture of Experts for Regime-Aware Factor Timing

A Reproducible Benchmark for Probabilistic Factor Allocation

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1. Problem Framing

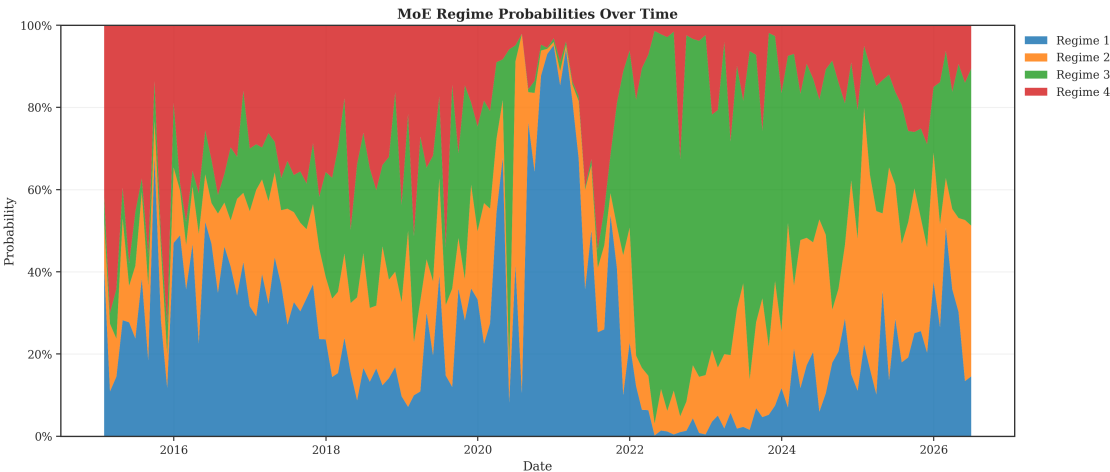
Equity factor premiums (Value, Momentum, Quality, Low Volatility) are not stable over time. They exhibit cyclicalities tied to changing macroeconomic conditions. Yet regimes are latent and unobservable—posing a challenge for dynamic allocation strategies.

Research Question: *Can a probabilistic model that explicitly represents uncertainty over latent economic regimes provide a coherent framework for out-of-sample factor allocation?*

Key Results (42 out-of-sample months):

Sharpe Ratio: **1.49** | Ann. Return: **40.61%** | Max DD: **-13.73%** | Win Rate: **69%**

Figure 1: Regime Uncertainty Over Time



MoE dynamically assigns probabilities to 4 latent economic regimes.

2. Methodology (Summary)

- **Data:** 155 months (2013–2026), 6 factors (SPY, IWD, MTUM, QUAL, USMV, VIX), FRED macro.
- **Features:** 96 lagged returns + FRED transformations.
- **Models:** Persistence, Rolling Avg, Momentum, Linear, RF, **MoE (K=4, EM, Ridge)**.
- **Backtest:** Expanding window, min_train=96, 10 bps costs.

3. Results Summary

The MoE model generated the highest Sharpe (1.49) and annualized return (40.61%).

Model	RMSE	Sharpe	Ann. Ret.	Max DD	Win Rate
MoE	33.68	1.49	40.61%	-13.73%	69%
Momentum	8.28	0.73	11.90%	-29.69%	62%
Rolling Avg	8.39	0.62	9.03%	-16.49%	64%
Linear	192.33	0.59	15.20%	-26.21%	48%

4. Limitations

This benchmark is based on 42 out-of-sample months of US equity data. Results are descriptive, not inferential, and do not constitute formal statistical evidence. The VIX spot index is not directly tradable, and FRED data are assumed to be available immediately.

5. Conclusion & Next Steps

This project provides an open-source, reproducible benchmark for evaluating probabilistic regime-aware models in factor timing. The full code, data pipeline, and complete paper are publicly available.

6. Data Sources

FRED: Federal Reserve Bank of St. Louis (fred.stlouisfed.org)

yfinance: Yahoo Finance historical data (finance.yahoo.com)

■ **Read the full paper & explore the code:**

github.com/kira-ml/mixture-of-experts-factor-timing